

Sanat Shourya

Data Analyst

Pennsylvania, USA | (214) 764-5371 | sanatshourya101@gmail.com | [LinkedIn](#)

SUMMARY

- Data Analyst with 3+ years of experience transforming healthcare and finance datasets into actionable insights using Python, SQL, Tableau, and Power BI on AWS and Azure, driving operational efficiency and data-driven BI improvements.
- Engineered ETL pipelines using Alteryx and SSIS to process large datasets from SQL Server, MySQL and AWS Redshift and established governance standards and automated validation frameworks for data integration and warehousing.
- Conducted statistical analysis and forecasting using Python (Pandas, SciPy) for hypothesis testing and A/B testing, and applied machine learning and data modeling to deliver insights guiding executive strategies in healthcare and finance operations.
- Developed and deployed Tableau and Power BI dashboards for KPI reporting and trend analysis, integrating cloud-based automation with real-time analytics to streamline reporting workflows and enhance stakeholder collaboration.

TECHNICAL SKILLS

Methodologies:	SDLC, Agile, Waterfall
Languages:	Python, R, SQL
IDEs:	Visual Studio Code, PyCharm, Jupyter Notebook
Packages:	NumPy, Pandas, Matplotlib, SciPy, Scikit-learn, TensorFlow, Seaborn, ggplot2
Visualization Tools:	Tableau, Power BI, Advanced Excel (Pivot Tables, XLOOKUP)
Cloud Platforms:	Amazon Web Services (AWS), Microsoft Azure
Databases:	MySQL, SQL Server, PostgreSQL, MongoDB, Oracle
Other Technical Skills:	SSIS, SSRS, Machine learning algorithms, ETL tools, EDA, Ad hoc Analysis, Statistics, Snowflake, Alteryx, Probability distributions, SAS, MS Visio, Hypothesis testing, Regression analysis, Linear Algebra, Advanced analytics, Data mining, Data visualization, Data warehousing, Data transformation, Data storytelling, Data integration, Data interpretation, Data pipeline, Association rules, A/B Testing, Forecasting and Modeling, Data cleaning, Data wrangling, Git, GitHub, Looker, QlikView, Microsoft Fabric, dbt, Collibra

EXPERIENCE

Data Analyst | CVS Health | USA

Aug 2024 – Present

- Built **ETL pipelines** using **Azure Data Factory**, **Synapse Analytics**, and **dbt** to integrate hospital billing data with insurance claims and patient records, streamlining reimbursement accuracy by **22%** and reducing claim denials.
- Developed automated daily data workflows using **Python** and **PostgreSQL** to ingest patient lab results, **EHR diagnostic records**, and pharmacy dispensing data, eliminating manual processing and accelerating clinical reporting cycles.
- Processed over **20 million records** from EHR encounter logs and medication administration reports using **Apache Spark**, boosting data readiness **4x** and ensuring timely availability for care coordination teams and regulatory submissions.
- Conducted **EDA** on readmissions/adverse-drug events/treatment outcomes using Agile sprints, identified seasonal trends and care-gap patterns that informed protocol updates and improved patient safety.
- Analyzed treatment protocol deviations and clinical outcomes using **SAS** and **SSIS**, enabling root cause quantification of complications and mortality variance, supporting data-driven adjustments to **HIPAA-compliant** care pathways.
- Performed **hypothesis testing** and **ANOVA** to validate intervention effectiveness, assess provider performance, identify admission trends, and support CMS-compliant quality metrics for strategic population health initiatives.
- Implemented **machine learning** anomaly detection using **logistic regression** and random forest in **Scikit-learn** to predict high-risk patient admissions, reducing false alerts and decreasing emergency response time by **30%**.
- Built clinical dashboards using **Advanced Excel**, **Pivot Tables**, **XLOOKUP**, and **Tableau** to visualize diagnosis trends by department and physician, replacing monthly reports and supporting quality improvement decisions during audits.

Data Analyst | Accenture | India

Apr 2021 – Jul 2023

- Created **Power BI dashboards** and **Excel** scorecards tracking portfolio efficiency, budget adherence, and financial KPIs across client segments and regions, enabling investment managers to act proactively before quarterly reviews.
- Automated ingestion, transformation, and validation of portfolio transactions and ledger data using **Alteryx workflows**, introducing reusable logic that reduced manual preparation efforts by **40%**, enhancing accuracy and efficiency.
- Applied **Python (SciPy, Statsmodels)** for **A/B testing** and time-series forecasting to evaluate expense controls and investment strategies, helping stakeholders prioritize initiatives that boosted operational efficiency by **25%**.
- Integrated unstructured compliance audit notes from **MongoDB** with structured Snowflake finance data, creating consolidated reporting layers for risk monitoring, cost allocation, and regulatory compliance tracking across teams.
- Validated spending patterns, detected anomalies, and supported **SOX** and **IFRS compliance** through **chi-square tests** and confidence intervals, ensuring accurate financial reviews across multiple business units.
- Executed financial data transformations using **Amazon Redshift on AWS**, centralizing key metrics and accelerating real-time **Power BI reporting** to enhance decision-making efficiency across portfolios.
- Led predictive analytical initiatives using **Python** and **TensorFlow** for clustering, regression, and classification, identifying risk contributors and underperforming portfolios, reducing financial losses and improving cash flow management.
- Collaborated with investment, finance, and audit teams to gather **BRDs** and **FRDs**, translating business rules into **SQL-driven** validation logic aligned with regulatory standards; facilitated **JAD** sessions for traceability.

EDUCATION

Master of Science in Computer Science

Aug 2023 – May 2025

Pennsylvania State University, University Park, USA

Bachelor of Technology in Computer Science

Aug 2019 – May 2023

Vellore Institute of Technology, Vellore, India

CERTIFICATIONS

- **Hacker Rank (Online Coding Platform)**: Rated 5-Star in SQL, C, C++, Problem Solving and Certified
- **Winner of Code2Create 5.0**: AHackathon by ACM (VIT Student Chapter)
- **Udemy/Coursera Certifications**: Business Analytics with Excel, The Complete 2023 Web Development Bootcamp, Algorithmic Thinking-II